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### Method of manufacturing diamond substrate

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#### Abstract

A method of manufacturing a diamond substrate includes: a step of placing a laser condensing unit **190** configured to condense laser light B so as to face an upper surface **10a** of a block **10** of single crystal diamond, a step of forming a modified layer **20**, which includes a processing mark **21** of graphite and a crack **22b** extending along a surface (111) around the processing mark **21**, in a partial region of the upper surface **10a** of the block **10** along the surface (111) of the single crystal diamond, along the surface (111) of the single crystal diamond at a predetermined depth from the upper surface **10a** of the block **10** by radiating the laser light B on the upper surface **10a** of the block **10** from the laser condensing unit **190** under predetermined conditions and condensing the laser light B inside the block **10**, and moving the laser condensing unit **190** and the block **10** in a relative manner two-dimensionally, and a step of forming a cleavage plane **25** at the predetermined depth of the remaining region of the upper surface **10a** of the block **10** by spontaneously propagating cleavage from the modified layer **20**.

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**Inventors:** Ikeno; Junichi (Saitama, JP), Yamada; Yohei (Saitama, JP), Suzuki; Hideki (Saitama, JP), Matsuo; Rika (Saitama, JP), Noguchi; Hitoshi (Tokyo, JP)

**Applicant:** Shin-Etsu Polymer Co., Ltd. (Tokyo, JP); Shin-Etsu Chemical Co., Ltd. (Tokyo, JP); National University Corporation Saitama University (Saitama, JP)

**Family ID:** 1000008821303

**Assignee:** Shin-Etsu Polymer Co., Ltd. (Tokyo, JP); Shin-Etsu Chemical Co., Ltd. (Tokyo, JP); National University Corporation Saitama University (Saitama, JP)

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*Primary Examiner:* Johnson; Christina A

*Assistant Examiner:* Liu; Xue H

*Attorney, Agent or Firm:* Nath, Goldberg & Meyer

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## Background/Summary

### CROSS REFERENCE TO RELATED APPLICATION

(1) This application is based upon and claims the benefit of priority under 35 U.S.C. § 119 from Japanese Patent Application No. 2021-410843 filed on Jul. 2, 2021, the entire contents of which are incorporated herein by reference.

### TECHNICAL FIELD

(2) The present invention relates to a method of manufacturing a diamond substrate, and more particularly to a method of manufacturing a diamond substrate by processing single crystal diamond using laser light.

### BACKGROUND

(3) Conventionally, silicon carbide (SiC) and gallium nitride (GaN) have been provided as semiconductor materials suitable for power devices instead of silicon (Si). However, diamond semiconductors have attracted attention as a next-generation material because they have a higher dielectric breakdown field, a higher power control index, and the highest thermal conductivity as compared with the above semiconductor materials, and research and development have been advancing toward practical application. Further, since a nitrogen-vacancy (NV) center in diamond is capable of highly sensitive magnetic detection at room temperature, this is expected to be applied to magnetic sensors, and research and development in this field have also been conducted (see WO 2015/107907 A1).

(4) Single crystal diamond which is expected to be applied to these semiconductors is synthesized by a high pressure and high temperature (HPHT) method and homoepitaxial growth, and these synthesis methods make it difficult to increase the area of a bulk substrate of single crystal diamond for use in semiconductor processes. Accordingly, a chemical vapor deposition (CVD) method has been applied in which single crystal diamond is heteroepitaxially grown by using a single crystal magnesium oxide (MgO) as a base crystal, because this CVD method is advantageous in increasing the area.

(5) In the heteroepitaxial growth by means of the CVD method, a bulk crystal of single crystal diamond, which is grown in the same orientation as the crystal orientation of the base MgO crystal, is obtained. That is, a bulk crystal of diamond with a crystal orientation [100] is obtained when the crystal orientation of the base MgO crystal is [100], and a bulk crystal of diamond with a crystal orientation [111] is obtained when the crystal orientation of the base MgO crystal is [111]. In the application of single crystal diamond to a magnetic sensor, it is necessary to form a high-density NV center and align the orientation axes of the NV center. Since a technique for orienting a high-density NV center in the [111] direction has been established by the CVD method, there has been an increasing need for a (111) bulk crystal using single crystal diamond having a main surface as a (111) surface (see WO 2015/046294 A1 and JP 2021-080153 A).

### SUMMARY

(6) Meanwhile, a bulk crystal of single crystal diamond obtained by heteroepitaxial growth is sliced and processed into a plate-like substrate, but diamond is hard and not easy to be processed. As a method for processing into a substrate, a smart cut technique is used in which a defective layer is introduced by ion implantation and is removed by etching to thereby perform delamination, but there has been a problem that a device in a high vacuum environment is required for ion implantation, and the processing time is long. Although it is possible to perform delamination with a thickness of several  $\mu\text{m}$ , there have been no cases of delamination with a thickness of several hundred  $\mu\text{m}$ .

(7) Other methods for processing into a substrate include polishing a bulk crystal of single crystal diamond, which is separated from a base crystal, to a desired thickness, or applying chemical

mechanical polishing (CMP). Further, single crystal diamond obtained by means of the conventional HPHT method is processed to slice an ingot, or a block made by further cutting an ingot to a fixed length, into a substrate, but there has been a problem that a loss occurs as a cutting margin. Since it is especially difficult to polish a bulk crystal of single crystal diamond with the [111] orientation, the development of a manufacturing method for obtaining a (111) substrate has been required.

(8) As described above, there has been a need for a manufacturing method in which a bulk crystal, an ingot, or a block of (111) single crystal diamond, which is expected to be applied to a high-precision magnetic sensor, is sliced into a substrate with a reduction in processing loss due to a cutting margin by means of a relatively simple method.

(9) The present invention has been made in view of the above circumstances, and an object of the present invention is to provide a method of manufacturing a diamond substrate by which a (111) substrate is manufactured with little processing loss from a bulk crystal of single crystal diamond with an [111] orientation, which is heteroepitaxially grown by a CVD method, and from an ingot or a block of single crystal diamond obtained by an HTHP method.

(10) In order to solve the above problem, a method of manufacturing a diamond substrate according to the present invention includes: a step of placing a laser condensing unit configured to condense laser light so as to face an upper surface of a block of single crystal diamond; a step of forming a modified layer, which includes a processing mark of graphite and a crack extending along a surface (111) around the processing mark, in a partial region of the upper surface of the block, along the surface (111) of the single crystal diamond, at a predetermined depth from the upper surface of the block by radiating the laser light on the upper surface of the block from the laser condensing unit and condensing the laser light inside the block, and moving the laser condensing unit and the block in a relative manner two-dimensionally; and a step of forming a cleavage plane at the predetermined depth of the remaining region of the upper surface of the block by spontaneously propagating cleavage from the modified layer.

(11) The method of manufacturing a diamond substrate may further include a step of causing the block to spontaneously delaminate into a portion up to a depth from the upper surface to the modified layer or the cleavage plane, and a portion deeper than the modified layer or the cleavage plane. The block may have a plate-like shape having an upper surface as the (111) surface of the single crystal diamond.

(12) The step of forming the modified layer may include a step of moving the laser condensing unit and the block in a relative manner in a predetermined scanning direction, and a step of moving the laser condensing unit and the block in a relative manner in a direction orthogonal to the scanning direction at a predetermined interval.

(13) The laser light may be pulsed laser light, and the graphite of the processing mark may be formed by laser light reflected by a crack extending from another adjacent processing mark in at least one of the scanning direction and the direction orthogonal to the scanning direction. The laser light may have a pulse width in a range of several ns to several hundred ns.

(14) The present invention makes it possible to manufacture a (111) substrate with little processing loss from a bulk crystal of single crystal diamond with an [111] orientation, and an ingot or a block of single crystal diamond obtained by an HTHP method, thereby making it possible to improve the yield in manufacturing the diamond substrate.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view illustrating a schematic configuration of a processing device.

(2) FIGS. 2A and 2B are perspective views illustrating a crystal structure of diamond.

- (3) FIG. 3 is a plan view of a block of single crystal diamond illustrating scanning of laser light.
- (4) FIGS. 4A to 4D are plan views illustrating the formation of a modified layer in the block of single crystal diamond.
- (5) FIGS. 5A to 5D are plan views illustrating the formation of a modified layer in the block of single crystal diamond.
- (6) FIG. 6 is a photograph illustrating a delaminated surface of the block of single crystal diamond which has delaminated at the modified layer or a cleavage surface.
- (7) FIGS. 7A and 7B are cross-sectional views illustrating a method of manufacturing a plurality of diamond substrates from the block of single crystal diamond.

#### DETAILED DESCRIPTION

- (8) Next, an embodiment of the present invention will be described with reference to the drawings. In the following drawings, the same or similar portions are denoted by the same or similar numerals. However, it should be noted that the drawings are schematic, and the relationship between the thickness and the plane dimensions, the ratios of the thickness of each layer, and the like are different from those in reality. Therefore, the specific thickness and dimensions should be determined in consideration of the following description. Also, it is needless to say that portions having different dimensional relationships and ratios among the drawings are included.
- (9) Further, the following embodiments exemplify devices and methods for embodying the technical idea of the present invention, and the embodiments of the present invention do not specify the materials, shapes, structures, arrangements, etc. of the components as follows. Various modifications may be made to the embodiments of the present invention in the claims.
- (10) FIG. 1 is a perspective view illustrating a schematic configuration of a processing device **100**. The processing device **100** includes a stage **110** for placing a block **10** of single crystal diamond thereon, a stage support unit **120** for supporting the stage **110** so as to be movable in the XY direction in a horizontal plane, and a fixture **130** for fixing the block **10** of single crystal diamond. An adhesive layer, a mechanical chuck, an electrostatic chuck, a vacuum chuck, or the like may be applied for the fixture **130**.
- (11) The plate-like block **10** having a rectangular outer periphery, which is formed by cutting an ingot of single crystal diamond to a predetermined length as an object to be processed, is fixed on the stage **110** with a (111) surface as an upper surface **10a**. The (111) surface serving as a main surface has an off-angle of 0 degrees. An object to be processed is not limited to this shape, and as long as the upper surface **10a** serves as the (111) surface, an ingot of single crystal diamond, a disk-shaped wafer, or a bulk crystal of single crystal diamond may be employed.
- (12) The processing device **100** includes a laser light source **160** for generating pulsed laser light, and a laser condensing unit **190** including an objective lens **170** and an aberration adjusting unit **180**, and radiates laser light B, which is emitted from the laser light source **160**, toward the (111) surface of the upper surface of the block **10** of single crystal diamond via the laser condensing unit **190**.
- (13) FIGS. 2A and 2B are perspective views illustrating a crystal structure of diamond. As illustrated in FIG. 2A, in a diamond crystal, carbon atoms indicated by white circles are covalently bonded to adjacent carbon elements by the arms of sp<sup>3</sup> hybrid orbitals extending in the direction of the four respective vertices of the tetrahedron centered on the carbon atoms. The covalent bonds are illustrated as solid lines. The carbon atoms, which are covalently bonded to each of the four adjacent carbon atoms, form a body-centered cubic lattice called a diamond structure.
- (14) FIG. 2B illustrates the (111) surface in the diamond structure. It is conventionally known that single crystal diamond is very hard because carbon atoms form covalent bonds with four respective adjacent carbon atoms in a diamond structure. However, a carbon atom is covalently bonded to an adjacent carbon atom in the <111> direction by only a single arm of the sp<sup>3</sup> hybrid orbital. Accordingly, when the covalent bond of the single arm is cut off, it is relatively easy to separate the

carbon atom from the adjacent carbon atom in the direction of the (111) surface orthogonal to the  $\langle 111 \rangle$  direction. As a result, the (111) surface becomes a cleavage surface.

(15) FIG. 3 is a plan view of the block **10** of single crystal diamond illustrating scanning of the laser light B. The block **10** of single crystal diamond placed on the stage **110** of the processing device **100** illustrated in FIG. 1 moves two-dimensionally in the XY direction in a horizontal plane relative to the laser condensing unit **190** such that the laser light B radiated from the laser light condensing unit **190** is radiated toward a predetermined position on the upper surface **10a** of the block **10** of single crystal diamond. Here, the block **10** has the (111) surface, which serves as a main surface with an off-angle of 0 degrees, as the upper surface **10a**.

(16) A scanning line **31** of the laser light B is first scanned in the [112] direction at a dot pitch  $dp$ , and shifted by a line pitch  $d$  in the [110] direction orthogonal to the [112] direction, and the laser light B is scanned in the [112] direction at a dot pitch  $dp$  to form a new scanning line **31**. By repeating the formation of the scanning line **31** in this way, a modified layer **20** is continuously formed along the (111) surface inside the block **10** of single crystal diamond.

(17) Inside the block **10** of single crystal diamond, the laser light B is condensed from the upper surface **10a** at a predetermined depth, and a processing mark of graphite and a crack extending along the (111) surface around the processing mark are formed. The laser light B of a nanosecond pulse laser, which is emitted from the laser light source **160**, is reflected by the crack formed along the (111) surface as the cleavage surface, and the diamond is pyrolyzed, and thus the processing mark of graphite is formed. Here, the nanosecond pulse laser refers to a laser having a pulse width, that is, a pulse duration in the range of several ns to several hundred ns, more specifically, in the range of 1 ns or more to less than 1  $\mu$ s.

(18) FIGS. 4A to 4D are plan views illustrating the formation of the modified layer **20** in the block **10** of single crystal diamond. FIGS. 5A to 5D are cross-sectional views illustrating the formation of the modified layer **20** in the block **10** of single crystal diamond. FIGS. 5A to 5D respectively correspond to cross sections formed by the cutting lines in FIGS. 4A to 4D.

(19) Referring to FIGS. 4A and 5A, a first scan line **31** is directed in the [112] direction, and a processing mark **21a** is directed toward the lower surface **10b** facing the upper surface **10a** in the block **10** of single crystal diamond. In addition, a crack **22a** extending in the [110] direction, and a crack **22bi** extending in the [110] direction are formed along the (111) surface due to cleavage from the processing mark **21a**, and the modified layer **20** including the processing mark **21a**, the crack **22a**, and the crack **22bi** is formed around the first scan line **31**. The processing mark **21a** has a conical shape which has the bottom surface near the cracks **22a** and **22bi** and the vertex in the direction of the lower surface **10b**. In the processing mark **21a**, the diamond is pyrolyzed into graphite by the condensed laser light B. Here, the length  $22biL$  of the crack **22bi** extending in the [110] direction is adjusted by the energy of the laser light B at the condensing point, the irradiated dot pitch  $dp$ , and the expansion of the processing mark **21a** controlled by the focal depth.

(20) Referring to FIGS. 4B and 5B, a second scan line **31** is performed after the laser condensing unit **190** moves in a relative manner two-dimensionally by the line pitch  $d$  from the first scan line **31** in the [110] direction orthogonal to the scanning direction of the first scan line **31**. At this time, the line pitch  $d$  is set such that the focal point of the laser light B is on the crack **22bi**. That is, the relationship between the line pitch  $d$  and the length  $22biL$  of the crack **22bi** is the length  $22biL > \text{the line pitch } d$ .

(21) A second scan line **31** is directed in the [112] direction, and a processing mark **21b** is directed toward the upper surface **10a**. In addition, a crack **22b** extending in the [110] direction along the (111) surface is formed due to cleavage from the processing mark **21b**, and the modified layer **20** including the processing mark **21b**, the cracks **22b** and **22bi** is formed around the second scan line **31**. The processing mark **21b** has a conical shape which has the bottom surface near the cracks **22b** and **22bi** and the vertex in the direction of the upper surface **10a**. In the processing mark **21b**, the diamond is pyrolyzed into graphite by the condensed laser light B.

(22) Here, cleavage of a length of  $22bL$  occurs in the crack  $22b$  along the (111) surface in the [110] direction due to expansion of the processing mark  $21b$ , and thus the modified layer  $20$  expands. At this time, cleavage also develops in the cracks  $22bi$  and  $22a$  due to the expansion of the processing mark  $21b$ , and as a result, a continuous cleavage surface is formed in the cracks  $22a$ ,  $22bi$  and  $22b$ .

(23) The scanning direction of the laser light B in the first scanning line  $31$  is not particularly limited to the [112] direction, and the scanning direction of the laser light B in the second scanning line  $31$  is not particularly limited to the [112] direction. The first scanning line  $31$  and the second scanning line  $31$  may be scanned in the [112] direction and in the [112] direction, respectively, or may be scanned in only one direction. In this regard, however, a scanning direction in which a reciprocal operation is performed is preferable in consideration of the efficiency of moving the laser condensing unit  $190$  and the block  $10$  of single crystal diamond in a relative manner.

(24) Referring to FIGS. 4C and 5C, a third scan line  $31$  is performed after the laser condensing unit  $190$  moves in a relative manner two-dimensionally by the line pitch  $d$  from the second scan line  $31$  in the [110] direction orthogonal to the scanning direction of the second scan line  $31$ . At this time, the line pitch  $d$  is set such that the focal point of the laser light B is on the crack  $22b$  where the cleavage has occurred. That is, the relationship between the line pitch  $d$  and the length  $22bL$  of the crack  $22b$  is the length  $22bL > \text{the line pitch } d$ .

(25) The third scan line  $31$  is directed in the [112] direction, and the processing mark  $21b$  is directed toward the upper surface  $10a$ . In addition, the crack  $22h$  extending in the [110] direction along the (111) surface is formed due to cleavage from the processing mark  $21b$ , and the modified layer  $20$  including the processing mark  $21b$  and the crack  $22b$  is formed around the third scan line  $31$ . The processing mark  $21b$  has a conical shape which has the bottom surface near the crack  $22bi$  and the vertex in the direction of the upper surface  $10a$ . In the processing mark  $21b$ , the diamond is pyrolyzed into graphite by the condensed laser light B. Here, cleavage of the length  $22bL$  occurs in the crack  $22b$  along the (111) surface in the [110] direction due to expansion of the processing mark  $21b$ , and thus the modified layer  $20$  expands.

(26) Referring to FIGS. 4D and 5D, the same operation is repeated after the fourth scan line  $31$  up to the  $n$ .sup.th scan line  $31$  toward the end face of the block  $10$ . When the effect of the graphitized processing mark  $21b$  in the  $n$ .sup.th scan line  $31$  exceeds a threshold value, cleavage develops from the processing mark  $21b$  in the  $n$ .sup.th scan line  $31$  to the end face of the block  $10$ . As a result, the modified layer  $20$  cleaved in the (111) surface along the crack  $22b$  is formed in the entire surface inside the block.

(27) A large amount of internal stress is accumulated in the modified layer  $20$  due to a rapid temperature change when forming the processing mark  $21b$  and the change in the crystal structure from diamond to graphite. In order to release such internal stress, the modified layer  $20$  spontaneously breaks along the (111) surface as the cleavage surface. Accordingly, at the modified layer  $20$ , the block  $10$  of single crystal diamond spontaneously delaminates into a first portion  $11$  from the upper surface  $10a$  to the modified layer  $20$ , and a second portion  $12$  from the modified layer  $20$  to the lower surface  $10b$ . At least one of the delaminated first portion  $11$  and the delaminated second portion  $12$  may be used as a substrate of single crystal diamond. This substrate may include a wafer.

(28) Here, most of the whole surface is taken up by the cleavage plane where the processing mark  $21b$  caused by the radiation of the laser light B does not exist, thereby making it possible to simplify the process of polishing a delaminated substrate, and obtain a delaminated surface having no damage caused by the radiation of the laser light B. Further, it is also possible to shorten the time required for scanning the scanning line  $31$ , thereby making it possible to greatly improve the productivity by shortening the processing time.

(29) In the block  $10$  of single crystal diamond delaminated into the first portion  $11$  and the second portion  $12$ , the modified layer  $20$  corresponds to a "cutting margin" to be lost by processing. The thickness of the modified layer  $20$  is equivalent to the height of the processing mark  $21b$  of

graphite having a substantially conical shape, and can be set within a range of several  $\mu\text{m}$  or less. Accordingly, it is possible to reduce the amount of single crystal diamond to be lost in manufacturing a diamond substrate by processing the block **10** of single crystal diamond, thereby making it possible to improve the yield in manufacturing the diamond substrate by processing the block **10** of single crystal diamond.

#### EXAMPLE

(30) In the processing device **100** illustrated in FIG. **1**, a nanosecond laser having the specification shown in TABLE 1 was used as the laser light source **160**. Further, as shown in TABLE 2, the block **10** of single crystal diamond was moved in a relative manner two-dimensionally with respect to the laser condensing unit **190**, and the laser light B was radiated toward the upper surface **10a**, which is the (111) surface, to form the modified layer **20** at a predetermined depth in a partial region of the upper surface **10a**. In this Example, Ib-type diamond obtained by means of the HPHT method was employed.

(31) TABLE-US-00001 TABLE 1 SPECIFICATION OF LASER LIGHT SOURCE LASER OSCILLATOR MODEL Hippo532-11 (MADE BY SPECTRA PHYSICS) WAVELENGTH 532 nm PULSE WIDTH 10 ns REPETITION FREQUENCY 20 kHz OUTPUT 1 W

(32) TABLE-US-00002 TABLE 2 LASER LIGHT RADIATION SETTINGS SCANNING VELOCITY 1 mm/s DOT PITCH  $d_p$  0.05  $\mu\text{m}$  LINE PITCH  $d$  30  $\mu\text{m}$

(33) The conditions for the laser light source **160** and the laser light radiation as above were set for the purpose of developing cleavage in the (111) surface due to the expansion of the processing mark **21b** by advancing the graphitization through the accumulation of the heat of the laser light B by the condensing unit, and increasing the density of the graphitization. In the above example, when the dot pitch  $d_p$  is set to 0.05  $\mu\text{m}$  and the line pitch  $d$  is set to 30  $\mu\text{m}$ , the graphitized processing mark is increased in density, and cleavage in the (111) surface develops due to stress in the crack **22b**. As a result, it is possible to perform cleavage in the (111) surface by performing laser radiation with ten scanning lines, that is, by forming the processing mark **21b** in a range of 300  $\mu\text{m}$  from the end of the diamond block.

(34) As for the size of the processing mark **21b**, it is necessary to control the processing mark **21b**, that is, the growth of the graphitized portion, in order to reduce the loss after delamination. The conditions were set considering that the depth (growth height) of the processing mark **21a** formed in the block **10** of single crystal diamond had to be set to 30  $\mu\text{m}$  or less at the maximum.

(35) As described above, the conditions obtained in consideration of the length of the crack **22b** and the size of the processing mark **21b** were that under a laser output of 1 W and an oscillation frequency of 20 kHz, a suitable range of the dot pitch  $d_p$  was 0.01  $\mu\text{m}$  to 0.5  $\mu\text{m}$ , and the range of the line pitch  $d$  at this time was 10  $\mu\text{m}$  to 50  $\mu\text{m}$ . Since it is difficult to measure the length of the crack **22b**, it is considered to be 100  $\mu\text{m}$  to 150  $\mu\text{m}$  in the range estimated from the above line pitch  $d$ . TABLEs 1 and 2 were set based on these considerations.

(36) Under the conditions as shown in TABLEs 1 and 2, a predetermined range of the upper surface **10a** of the block **10** of single crystal diamond in the vertical direction was scanned by using the ten scanning lines **31**, and the modified layer **20** was formed in this scanned region at a predetermined depth from the upper surface **10a**. As a result, the cleavage propagated spontaneously along the (111) surface of the cleavage surface at the same predetermined depth as that of the modified layer **20** starting from the modified layer **20**, and a cleavage plane **25** was formed at the predetermined depth over the entire surface of the remaining region excluding the partial region where the modified layer **20** was formed. Further, the block **10** of single crystal diamond spontaneously delaminates at the modified layer **20** or the cleavage plane **25** to obtain the first portion **11** from the upper surface **10a** to the modified layer **20** or the cleavage plane **25**, and the second portion **12** deeper than the modified layer **20** or the cleavage plane **25**.

(37) FIG. **6** is a photograph illustrating a delaminated surface of the block **10** of single crystal diamond which has delaminated at the modified layer **20** or the cleavage plane **25**. In the



delaminated surface, the processing marks **21** using graphite formed along the ten scanning lines **31** are observed. It is observed that the cleavage plane **25**, which is the (111) surface, forms a smooth surface in the portion other than the processing marks **21** in the delaminated surface.

(38) In the above example, when the dot pitch  $dp$  is set to  $0.05\ \mu\text{m}$  and the line pitch  $d$  is set to  $30\ \mu\text{m}$ , the graphitized processing marks are increased in density, which results in cleavage in the (111) surface developing due to stress in the crack **22b**. As a result, it is possible to perform cleavage in the (111) surface by performing laser radiation with the ten scanning lines, that is, by forming the processing mark **21b** in a range of  $300\ \mu\text{m}$  from the end of the diamond block. Since the outer dimension of the block **10** of single crystal diamond used in this example in the [110] direction is  $2.5\ \text{mm}$ , about 90% of the surface of the diamond substrate obtained by delamination is formed by the cleavage plane in the (111) surface where the processing mark **21b** does not exist. As the cleavage plane obtained in this way is a smooth surface, there is a great advantage that the next processing step such as polishing and etching can be greatly simplified.

(39) FIGS. 7A and 7B are cross-sectional views illustrating a method of manufacturing a plurality of diamond substrates from the block **10** of single crystal diamond. FIG. 7A illustrates the block **10** of single crystal diamond in which the modified layer **20** is formed before delamination, and FIG. 7B illustrates the second portion **12** where the modified layer **20**, which is a cleavage surface, has delaminated from the block **10** of single crystal diamond. The upper surface **12a** of the second portion **12** corresponds to the cleavage plane in the block **10** of single crystal diamond, and the lower surface **12b** of the second portion **12** corresponds to the lower surface **10b** of the block **10** of single crystal diamond.

(40) Radiating the laser light B from the upper surface **12a** of the second portion **12** in accordance with TABLES 1 and 2, forming the modified layer **20** in the second portion **12**, and delaminating the modified layer **20** as a cleavage surface make it possible to manufacture a new diamond substrate. Repeating this process brings the great advantage that it is possible to produce a substrate from diamond, which is a difficult-to-cut material, by means of an extremely efficient and simple method with greatly reduced processing loss.

(41) The present invention can be utilized for manufacturing a power device or a magnetic sensor using a diamond substrate.

## Claims

1. A method of manufacturing a diamond substrate comprising: a step of placing a laser condensing unit configured to condense laser light so as to face an upper surface of a block of single crystal diamond; a step of forming a modified layer, which includes a processing mark of graphite and a crack extending along a surface (111) around the processing mark, in a partial region of the upper surface of the block along the surface (111) of the single crystal diamond, at a predetermined depth from the upper surface of the block by radiating the laser light on the upper surface of the block from the laser condensing unit and condensing the laser light inside the block, and moving the laser condensing unit and the block in a relative manner two-dimensionally, wherein the step of forming the modified layer includes: a step of moving the laser condensing unit and the block in a relative manner in a predetermined scanning direction; and a step of moving the laser condensing unit and the block in a relative manner in a direction orthogonal to the scanning direction at a predetermined interval, wherein the laser condensing unit moves in a relative manner two-dimensionally by a predetermined line pitch in the direction orthogonal to the scanning direction; a step of forming a cleavage plane at the predetermined depth of a remaining region of the upper surface of the block by spontaneously propagating cleavage from the modified layer; and a step of causing the block to spontaneously delaminate into a portion up to a depth from the upper surface to the modified layer or the cleavage plane, and a portion deeper than the modified layer or the cleavage plane, wherein the block has a planar surface as an upper surface as the (111) surface of the single crystal

diamond.

2. The method of manufacturing a diamond substrate according to claim 1, wherein the laser light is pulsed laser light, and the graphite of the processing mark is formed by laser light reflected by a crack extending from another adjacent processing mark in at least one of the scanning direction and the direction orthogonal to the scanning direction.
  3. The method of manufacturing a diamond substrate according to claim 1, wherein the laser light has a pulse width in a range of several ns to several hundred ns.
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